

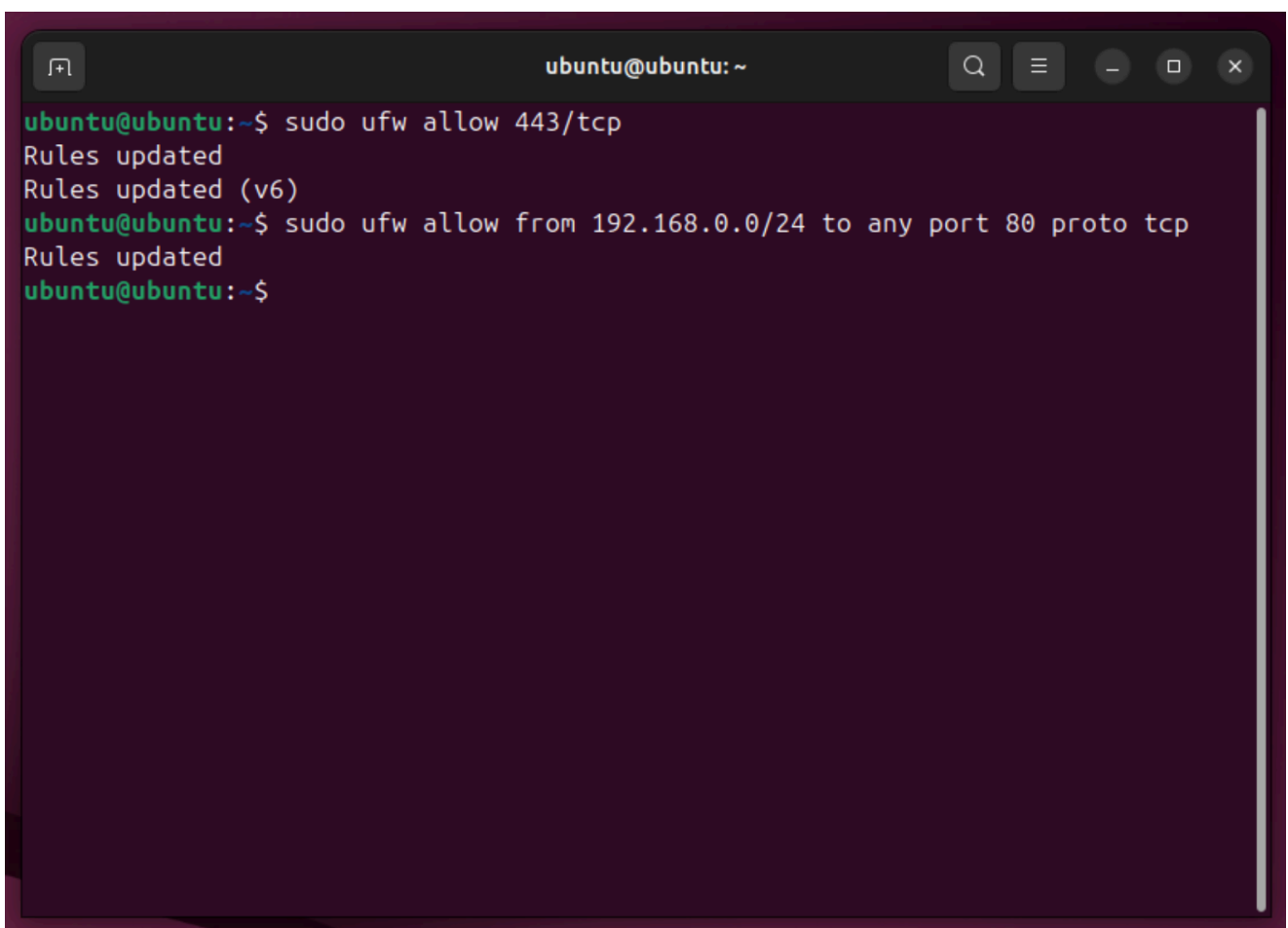
OS and Network Security

1 - Configure Firewall

Using `ufw`, configure the firewall to allow https from any IP address, but http from only the following subnet: 192.168.0.0/24

Submit a screenshot of your firewall rules.

```
ufw allow 443/tcp
ufw allow from 192.168.0.0/24 to any port 80 proto tcp
```

A terminal window titled 'ubuntu@ubuntu: ~' with search, menu, and window control icons in the title bar. The terminal shows the execution of two 'ufw' commands. The first command is 'sudo ufw allow 443/tcp', which results in 'Rules updated' and 'Rules updated (v6)'. The second command is 'sudo ufw allow from 192.168.0.0/24 to any port 80 proto tcp', which results in 'Rules updated'. The prompt returns to 'ubuntu@ubuntu:~\$' after each command.

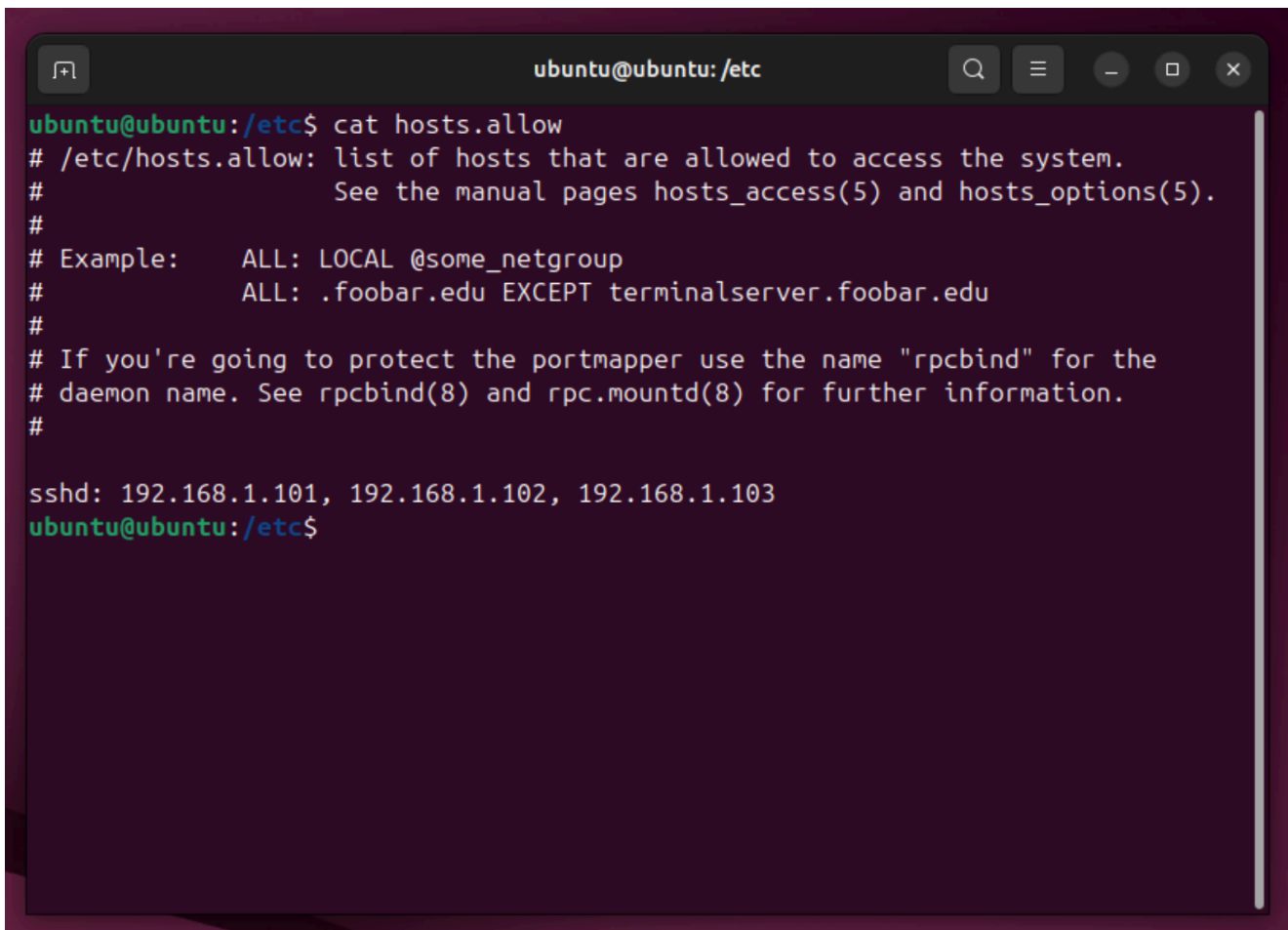
```
ubuntu@ubuntu:~$ sudo ufw allow 443/tcp
Rules updated
Rules updated (v6)
ubuntu@ubuntu:~$ sudo ufw allow from 192.168.0.0/24 to any port 80 proto tcp
Rules updated
ubuntu@ubuntu:~$
```

2 - Configure SSH Host Access

Configure the server to only allow the following 3 IPs to login to the server. Please note, this change should be made in `hosts.allow`:

- 192.168.1.101
- 192.168.1.102
- 192.168.1.103

Submit a screenshot of the configuration file you modified by using the `cat` command to show the contents of the file.

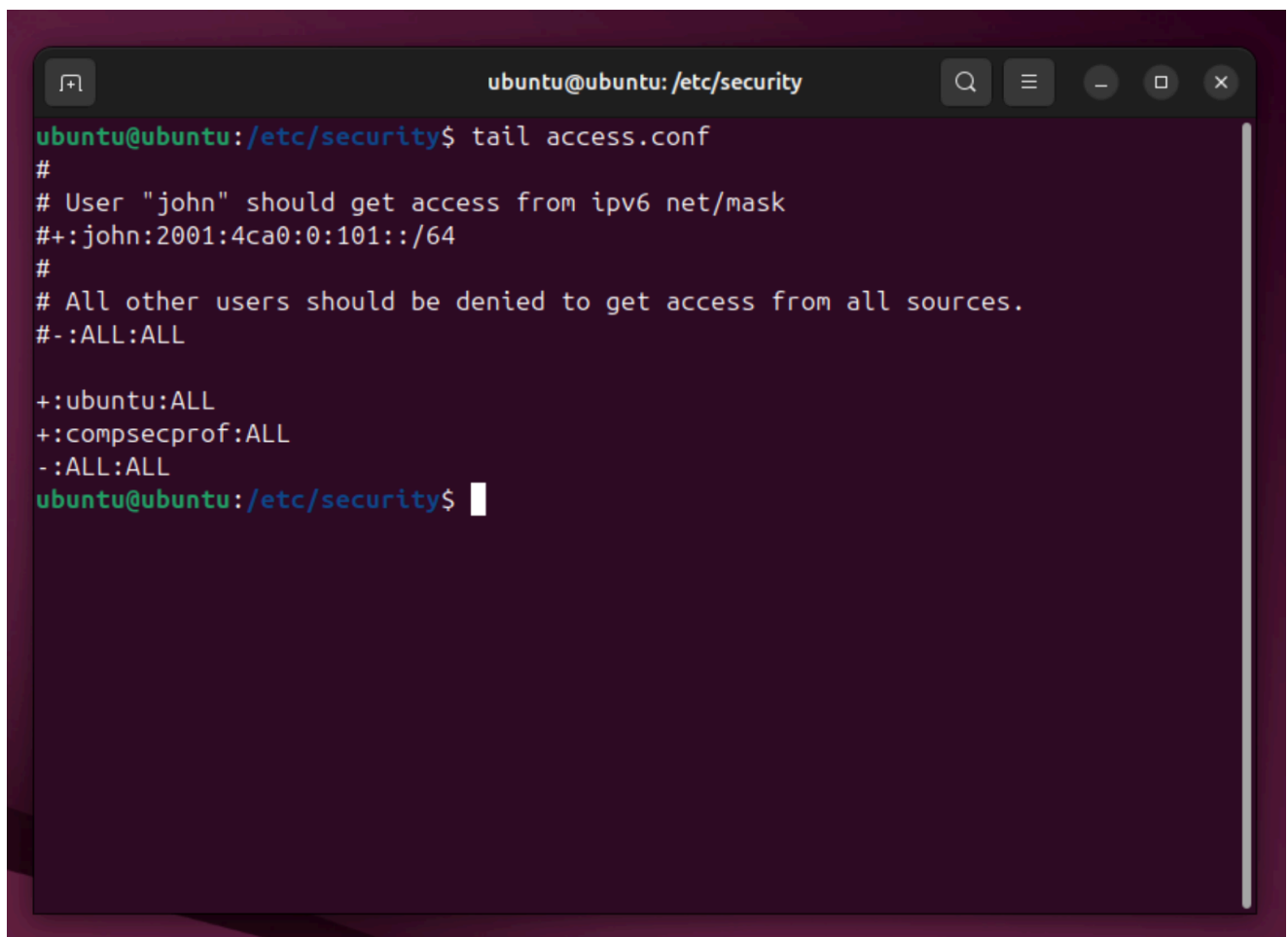
A terminal window titled 'ubuntu@ubuntu: /etc' with search, menu, and window control icons. The command 'cat hosts.allow' has been executed, displaying the following text:

```
ubuntu@ubuntu:/etc$ cat hosts.allow
# /etc/hosts.allow: list of hosts that are allowed to access the system.
#           See the manual pages hosts_access(5) and hosts_options(5).
#
# Example:   ALL: LOCAL @some_netgroup
#           ALL: .foobar.edu EXCEPT terminalserver.foobar.edu
#
# If you're going to protect the portmapper use the name "rpcbind" for the
# daemon name. See rpcbind(8) and rpc.mountd(8) for further information.
#
sshd: 192.168.1.101, 192.168.1.102, 192.168.1.103
ubuntu@ubuntu:/etc$
```

3 - Configure Server Username Access

Configure the server to only allow the following accounts to sign in. Please note. This change should be made in `access.conf`:

- Your username for the system and a new user named "compsecprof"
- Submit a screenshot of the configuration file you modified by using the `tail` command to show the contents of the file.

A terminal window with a dark purple background. The title bar at the top reads 'ubuntu@ubuntu: /etc/security'. The terminal shows the command 'tail access.conf' being executed. The output displays the contents of the file, which includes comments and access rules for users like 'john', 'ubuntu', and 'compsecprof'. The prompt 'ubuntu@ubuntu: /etc/security\$' is visible at the bottom.

```
ubuntu@ubuntu: /etc/security$ tail access.conf
#
# User "john" should get access from ipv6 net/mask
#+:john:2001:4ca0:0:101::/64
#
# All other users should be denied to get access from all sources.
#-:ALL:ALL

+:ubuntu:ALL
+:compsecprof:ALL
-:ALL:ALL
ubuntu@ubuntu: /etc/security$
```

4 - Perform a NMAP Scan

Perform a nmap scan of `scanme.nmap.org` where you can determine the version of software running on the server.

Submit a screenshot of the command and the result.

```
ubuntu@ubuntu: /etc/security$ sudo nmap -sV --version-all -O -T4 scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-06 04:32 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.065s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
2000/tcp  open  tcpwrapped
5060/tcp  open  tcpwrapped
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Aggressive OS guesses: IPFire 2.27 (Linux 5.15 - 6.1) (95%), Linux 4.19 - 5.15 (95%), Linux 4.15 (94%), Linux 5.4 (94%), Linux 5.18 (91%), Android TV OS 11 (Linux 4.19) (91%), IPFire 2.25 firewall (Linux 4.14) (91%), Linux 2.6.32 (91%), Linux 2.6.39 (91%), Linux 3.10 - 3.12 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 23 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.27 seconds
ubuntu@ubuntu: /etc/security$
```

5 - Configure SSHD

Configure SSHD to not allow password based authentication and to not allow root to directly login to the server. Also only allow 2 active sessions at a time and 2 max authentication tries. This change should be made within the SSH configuration file.

Submit a screenshot of the changes within vi or nano. Please be sure it includes all changes.

```
ubuntu@ubuntu: /etc/ssh$ cat sshd_config
PasswordAuthentication no
PermitRootLogin no
MaxSessions 2
MaxAuthTries 2
ubuntu@ubuntu: /etc/ssh$
```